ModelIt!

Teacher's Guide

Can Plants Grow Without Soil?

What Plants Really Need to Survive

Quick Reference

Grade Level	Grade 2
NGSS Standard	2-LS2-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: 2-LS2-1

Plan and conduct an investigation to determine if plants need sunlight and water to grow.

Three-Dimensional Learning

Science Practice	**Planning and Carrying Out Investigations** — Students plan and conduct an investigation to determine if plants need sunlight and water to grow, testing different conditions.
Disciplinary Core Idea	**LS2.A Interdependent Relationships in Ecosystems** — Plants depend on water and light to grow. Plants depend on animals for pollination or to move their seeds around.
Crosscutting Concept	**Cause and Effect** — Students observe that removing sunlight or water causes poor plant growth, establishing a cause-and-effect relationship.

CAST Testing Alignment

MULTIPLE CHOICE:

A student puts one plant in sunlight with water and another plant in a dark closet with water. After two weeks, the plant in the dark is pale and droopy. What does this tell us?

CONSTRUCTED RESPONSE:

Explain what plants really need to grow. Can a plant grow without soil? Use evidence from your investigation to support your answer.

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Plants are amazing living things that make their own food from sunlight, water, and air. Understanding what plants truly need to grow helps us grow food in new and creative ways — even in places without soil!

What Do Plants Need?

Plants need four things to survive and grow: sunlight, water, air (specifically carbon dioxide), and nutrients. Sunlight gives plants the energy to make food in their leaves through a process called photosynthesis. Water travels up through the roots and into every part of the plant. Nutrients are like vitamins that help the plant grow strong.

The Role of Soil

Soil is helpful for plants, but it is not actually required. Soil holds water near the roots so the plant can drink. It also contains nutrients that the plant needs. But if we give the plant water and nutrients a different way, the plant does not need soil at all. This is the idea behind hydroponics — growing plants in water.

Plants in Space

NASA scientists grow plants on the International Space Station without any soil! They use special systems that deliver water, nutrients, and light directly to the roots. Astronauts have grown lettuce, radishes, and peppers in space. This research helps us figure out how to grow food for long missions to the Moon and Mars.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify sunlight amount and water availability as external components and plant growth as an internal component that depends on both sunlight and water.

E **Establish Relationships**
Students determine that both sunlight and water have positive effects on plant growth — more of each helps the plant grow taller and healthier.

V **Visualize & Model**
Students build a model showing how sunlight and water together determine how well a plant grows.

E **Evaluate Outcomes**
Students run full sun/water, no sunlight, and no water scenarios to see the effects of removing each need.

R **Revise & Extend**
Students add nutrients in water or air flow to explore other factors that affect plant growth.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"Do you think a plant can grow without any dirt? Just water and sunshine? Let us find out!"*

Watch For

Show a plant growing in a clear jar of water. Let students observe the roots.

Slide 2: What Will We Learn Today?

3 min

Say: *"Today we are going to investigate what plants REALLY need to grow."*

Watch For

Read objectives aloud. Introduce nutrients, sunlight, roots, and germinate with real examples.

Slide 3: The Big Question

3 min

Say: *"Everybody thinks plants need soil. But what if they do not? What do plants REALLY need?"*

Watch For

Quick poll: Raise your hand if you think plants MUST have soil. We will find out who is right!

Slide 4: Let's Build a Model!

3 min

Say: *"We are going to build a model to test what happens when plants get different things."*

Watch For

Preview LEVER steps. Show examples of plants in soil and plants in water side by side.

Slide 5: Activity 1: Sort the Parts

7 min

Say: *"What are the parts of our plant system? Sunlight and water are inputs we can control. Plant growth is what we observe."*

Watch For

Guide sorting: Sunlight and Water are external, Plant Growth is internal.

Slide 6: Activity 2: Connect with Arrows

7 min

Say: *"When we give a plant MORE sunlight, does it grow better or worse? What about more water?"*

Watch For

Draw positive arrows from Sunlight and Water to Plant Growth. Discuss why both are needed.

Slide 7: Activity 3: Run the Simulation!

8 min

Say: *"Let us test three conditions: lots of sun and water, no sun, and no water. Predict what happens!"*

Watch For

Students predict, then run simulations. Compare results. Discuss which conditions matter most.

Slide 8: What Did We Discover?

7 min

Say: *"So do plants need soil? The answer might surprise you — they need what is IN the soil, not the soil itself!"*

Watch For

Show a hydroponic plant as proof. Discuss how astronauts grow food in space without soil.

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Sunlight Amount, Water Availability
- **INTERNAL:** Plant Growth

Relationships

- **Sunlight Amount to Plant Growth:** POSITIVE (+) — More sunlight gives the plant more energy to make food in its leaves. Plants in bright sunlight grow taller and greener. Plants with no sunlight become pale and weak because they cannot make food.
- **Water Availability to Plant Growth:** POSITIVE (+) — More water available to the roots means the plant can stay hydrated and transport nutrients. Plants with plenty of water grow well. Plants without water wilt and eventually stop growing.

Key Discoveries

- Plants need BOTH sunlight and water to grow — missing either one causes the plant to struggle
- Plants can grow without soil if they get water and nutrients another way — this is called hydroponics
- Sunlight gives plants energy to make their own food inside their leaves
- Roots soak up water and nutrients, whether they are in soil or in water

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Plants eat soil"	Plants do NOT eat soil! Plants make their own food using sunlight, water, and air. Roots absorb water and tiny nutrients from the soil, but the soil itself is not food. That is why plants can grow in just water if nutrients are added.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Plants only need water to survive"	Water is very important, but it is not enough by itself. Plants also need sunlight to make food and air to breathe. A plant in a dark room with plenty of water will still get sick and pale because it cannot make food without light.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Plants get their food from soil or fertilizer"	Fertilizer and soil nutrients are like vitamins — they help the plant grow strong, but they are not the plant's main food. Plants make their own food from sunlight and air through photosynthesis. Nutrients from soil are just helpers.